Ellan Vannin Wealth Management

Strategic Infrastructure Investment Proposal

The Midland High-Speed Upgrade Programme

London – Leicester – Nottingham – Derby – Sheffield

A 125–160mph East Midlands Main Line Modernisation Partnership

Prepared by:
Ellan Vannin Wealth Management (EVWM)
Infrastructure & Strategic Assets Division

Executive Summary

Ellan Vannin Wealth Management proposes a long-term Public Private Infrastructure Partnership with the British Government to transform the **Midland Main Line** into a high-speed intercity railway comparable in performance to HS2, but delivered through targeted upgrades rather than an entirely new railway.

The proposal:

The Midland High-Speed Upgrade (MHSU)

would upgrade the existing corridor:

London St Pancras
↓
Luton
↓
Bedford
↓
Kettering
↓
Market Harborough
↓
Leicester
↓
Derby / Nottingham

↓
Chesterfield
↓
Sheffield

into a high-capacity, electrified, high-speed railway.

The objective:

> Deliver HS2-style journey times on an existing strategic corridor at a fraction of the cost of a new-build high-speed line.

The Midland Main Line is already undergoing major modernisation, including electrification works, and serves London, Leicester, Nottingham, Derby and Sheffield.

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1. Policy Rationale

The Problem

Britain has historically invested in new railways rather than maximising existing corridors.

HS2 demonstrates the benefits of dedicated high-speed infrastructure, but its cost has increased substantially, with current estimates exceeding £87bn–£100bn.

EVWM proposes an alternative:

"HS2 Performance Through Strategic Engineering"

Instead of:

- hundreds of kilometres of new railway
- expensive urban tunnels
- entirely new stations

upgrade Britain's existing Midland corridor.

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2. Strategic Vision

The project creates:

The Midlands Economic Spine

Connecting:

London

Financial centre

East Midlands

Manufacturing, logistics, universities

South Yorkshire

Advanced manufacturing and engineering

Economic corridor:

Population connected:

15+ million people

Economic output:

£300bn+ regional economy

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3. Proposed Infrastructure Works

Phase One

London – Bedford High-Speed Upgrade

Current issues:

- commuter congestion
- mixed traffic
- slow sections

Works:

- four tracking where required
- junction separation
- grade-separated crossings
- tunnel improvements
- signalling upgrade

Bedford Fast Rail Tunnel

A major engineering intervention.

Proposal:

New twin-bore tunnel bypassing the slowest sections around Bedford.

Purpose:

- remove commuter conflicts
- allow 125–160mph running
- separate regional and express services

Phase Two

Bedford – Leicester High-Speed Corridor

Works:

- straighten curves
- bypass slow sections
- rebuild junctions
- additional tracks
- electrification

Major intervention:

Market Harborough Speed Tunnel

A new alignment bypassing existing restrictive curves.

Target:

London → Leicester

Current:

approximately 1h05–1h15

Future:

45 minutes

Phase Three

Leicester Northern Tunnel

Leicester is one of Britain's fastest-growing cities but suffers from railway constraints.

Proposal:

A new underground railway section:

- bypassing the existing constrained station throat
- allowing through running
- increasing capacity

Similar principle:

Crossrail-style urban railway engineering.

Phase Four

Leicester – Sheffield Upgrade

Route:

Leicester

↓

East Midlands Parkway

↓

Derby

↓

Chesterfield

↓

Sheffield

Works:

- four tracking sections
- curve improvements
- electrification
- tunnels through difficult terrain

Peak District Engineering Challenge

The existing Midland Main Line faces geographical constraints approaching Sheffield.

Proposal:

New Peak District Rail Tunnel

A strategic tunnel:

Approximate length:

15–20km

Purpose:

- reduce gradients
- remove slow curves
- enable high-speed operation

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4. Railway Specification

Feature	Specification
Electrification	25kV AC
Maximum speed	125–160mph
Signalling	ETCS Level 2/3
Track	Four-track sections
Rolling stock	High-speed electric trains
Freight	Dedicated capacity

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5. Journey Time Revolution

Current Approximate Times

Journey	Current
London–Leicester	~1h05
London–Nottingham	~1h40
London–Derby	~1h35
London–Sheffield	~2h00

Proposed HS2-Level Upgrade

Journey	Future
London–Leicester	**45 min**
London–Nottingham	**65 min**
London–Derby	**70 min**
London–Sheffield	**85 min**

Wider Connections

Route	Future Time
Sheffield–London	1h25
Nottingham–London	1h05
Derby–London	1h10

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6. Economic Impact

The East Midlands Transformation

The region has historically underperformed compared with its population and industrial potential.

A faster railway creates:

- larger labour markets
- inward investment
- university connections
- business travel
- housing growth

GDP Impact Model

Connected economy:

Approximately:

£300bn

Assumed productivity improvement:

1%

Economic uplift:

£3 billion annually

Additional benefits:

Sector	Impact
Business investment	£1bn/year
Housing development	£500m/year

Sector	Impact
Tourism	£300m/year
Labour productivity	£1.5bn/year

Total Potential Economic Benefit

£3–5 billion annually

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7. Capital Cost Estimate

A targeted upgrade approach:

Component	Cost
Electrification completion	£2bn
Track upgrades	£4bn
Bedford tunnel	£2.5bn
Leicester improvements	£2bn
Peak District tunnel	£5bn
Stations/signalling	£2bn

Total Programme Cost

£17–20 billion

Compared with:

HS2:

£87–100bn+ programme estimates.

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8. Financing Structure

Midland Infrastructure Partnership

Created by:

- British Government
- Ellan Vannin Wealth Management

- UK pension funds
- insurance investors
- infrastructure funds

PFI Structure

Government Provides:

- strategic guarantee
- planning support
- availability payments

EVWM Provides:

- capital raising
- project management
- private investment
- asset management

Revenue Sources

Passenger Services

- premium intercity fares
- increased passenger volumes

Property Development

Around:

- Leicester
- Nottingham
- Derby
- Sheffield

Freight Access Charges

Additional capacity creates freight opportunities.

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9. Investment Returns

Infrastructure characteristics:

- 100-year asset life
- inflation-linked revenues
- strategic importance

Target:

6–8% infrastructure return

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10. Strategic Recommendation

Ellan Vannin Wealth Management recommends that the British Government establish:

The Midland High-Speed Upgrade Partnership

as an alternative strategic rail programme.

The objective:

Create a railway with:

- HS2-level journey times
- lower construction cost
- faster delivery
- greater regional coverage

Final Investment Thesis

Britain does not only need new railways.

It needs better use of the railways it already has.

The Midland Main Line is one of Britain's most important economic corridors.

By combining:

- targeted tunnels
- electrification
- bypasses
- capacity expansion
- digital signalling

the UK can create:

A 21st-century Midland High-Speed Railway

connecting London, the East Midlands and South Yorkshire in a way that unlocks billions in additional economic growth while avoiding the full cost of a completely new HS2-style route.